

Quiz #5 (Due 2019/3/22) 23:59 p.m. (4% + 1% Bonus)

Name: Zhuang Liu

CompSci234/NetSys210 Advanced Topics in Networking

1) (1%) **Decentralized Social Media.** In this paper, which of the following aspect is not considered by the replica node selection algorithms?

- a) Online time
- b) Storage capacity B
- c) User experience
- d) Social relation

2) (1%) **Information Centric Networks.** What are the three main components that make up an NDN?

- 1. A pending interest table (PIT)
- 2. A forwarding information base (FIB)
- 3. A content store (CS)

3) (1%) **Media Aware Network Element.** Please list at least one advantage of using an OpenFlow controller in the proposed controlled multicast.

The capability of the terminals can be collected

It is able to acquire the network conditions

Provides admission control in a multicast context

Provide in-network adaptation

Support heterogeneous devices with different display capability

- 4) (1%) **OpenNebula Distributed Cloud.** How does OpenNebula solve the problems caused by “churn” in a volunteer-based environment?

Resource Manager use a dynamic lease mechanism to help master nodes to compute the estimated time before assigning the volunteer nodes, thus ensure fault tolerance.

If churn happens after the assignment then replication will solve the problem. The master node would have copies of either the data or the computation and will resort to them.

- 5) (1%) **Energy Efficient Internet-of-Things.** What does “out-of-band mechanism for security” mean? Please give an example.

OOB Mechanism: Utilizing two separate networks or channels, one of which is different from the primary network or channel, to simultaneously communicate between two parties or devices for identifying a user.

Example: A cellular network is commonly used for OOB authentication. An example of OOB authentication is when an online-banking user is accessing her online bank account with a login and a one-time password (OTP) is sent to her mobile phone via SMS (short message service).